

SPARK-2024: Image detection and Segmentation

Nabeel Ahsan
0240764054
nabeel.ahsan001@uni.lu

Mo Nafees
0242805924
nafees.mo001@uni.lu

Arooba Arshad
023063191F
arooba.arshad.001@student.uni.lu

Raja Venkat Sai Snehith
0240260900
0240260900@uni.lu

Abstract—This project aims to address the challenge of identifying spacecraft in images collected from orbit by utilizing two complementary tasks - object detection and semantic segmentation. Object detection utilizes a transformer-based detector to identify spacecraft belonging to different mission classes in an instance-wise manner. Semantic segmentation employs a convolutional neural network for pixel-wise classification of images containing the spacecraft and its components (i.e. separating the spacecraft from the background, the spacecraft's body, and the solar panels). Object detection performance is measured using COCO-style metrics, including a precision-recall analysis, class-wise confusion analysis based on the bounding box Intersecting Over Union (IoU) threshold for the spacecraft, and segmentation performance is measured by pixel-wise confusion matrices and pixel-wise IoU. Results from experiments conducted on the SPARK-2024 validation set demonstrate near perfect detection accuracy when detecting all types of spacecrafts and provide evidence of excellent background segmentation performance while demonstrating ongoing confusion regarding some fine grained component classifications. This information sheds light on the current advantages and limitations of deep learning methodologies for visual understanding from space, thus creating a foundation for continued improvements on a component-level for classification and segmentation.

Index Terms—COCO, RT-DETRv2, DeepLabV3, segmentation, classification

I. INTRODUCTION

As the field of computer vision is developing and progressing rapidly, artificial intelligence (AI) driven systems have considerably improved their ability to understand and analyze visual data. The key objectives of this field are object detection and semantic segmentation. These two domains are critical for numerous applications, including autonomous systems, remote sensing, medical imaging, and monitoring of space missions. In satellite imaging systems, reliable detection and segmentation of objects is critical to the success of space situational awareness, spacecraft monitoring, and mission analysis.

The SPARK-2024 spacecraft dataset offers a rigorous benchmark for spacecraft perception, featuring authentic satellite images of many spacecraft classifications in differing visual conditions. The dataset serves as a catalyst for building strong, high-performing deep learning pipelines that handle large volumes of data while achieving the highest levels of accuracy.

The goal of this project is to develop, implement, and evaluate a fully operational deep learning pipeline for spacecraft detection and segmentation. For the detection task, the project employs RT-DETRv2 model, which uses transformer

architecture with convolutional extraction methods and transformer encoders, has been adopted to provide users with the capability of fast and precise object detection. To perform pixel-level segmentation of the spacecraft's body and solar panel region, a separate model has been developed specifically for segmentation. Both detection and segmentation tasks have been compiled into a single repeatable/highly reproducible pipeline hosted on an HPC system.

The pipeline we put together starts with getting the data ready. That means preparing datasets and turning them into the COCO format, which is Common in this Context. Then there's model training, using configurable YAML files to set up experiments. After that, we use test datasets to run large-scale inference. After the model training phase, the final step is to utilize a high-performance computing environment that has been configured to support GPU acceleration through SLURM to run large-scale inferences on test datasets.

II. RELATED WORK

In [1] Tseng et. al. have proposed an architecture that uses both segmentation and object detection in a single unified network. The architecture is made up of a base network, a segmentation decoder and a detection decoder. With this they were trying to build a single model that is able to detect cars and people on roads. The model was tested on a simulation built using Unity. The base network was composed of Single Shot Detector (SSD) layers. The segmentation decoder is made up of 7 transposed convolution layers with the depth of the last layer being 4 for the 4 classes - people, cars, road and background. For the detection decoder, they used the 4 feature layers from the encoder and added a convolution layer after each feature layer. Their unified model was able to yield 99.7% accuracy for segmentation tasks and 99.45% for detection tasks. This paper aligns closely with our project, since both combine detection and segmentation and evaluate performance on rendered images/environments.

Chen et. al. try to address the problem of waste management, specifically construction waste in [2] using DeepLabv3+. They used a dataset of 5,366 images and tried to identify 10 different wastes, including, bricks, rock, plastic, but also, the truck carrying the waste itself. After cleaning the dataset, 5022 viable samples were found fit for their task. For the backbone network of the encoder, two networks were chosen - Xception and ResNet. This is then fed into atrous spatial pyramid pooling (ASPP) module which extracts features using different rates r_i which are then concatenated

with the result of 1×1 convolution and the global average pooling feature to achieve scale invariance. The proposed approach achieved a mean Intersection over Union (mIoU) of 0.56 in segmenting nine types of materials/objects with a time performance of 0.51 s per image.

Wang et. al in their research [3] present model called Group DETRv2 built on a ViT-Huge encoder and the decoder was a combination of Group DETR [4] and DINO [5]. Training was done on A100 GPUs was able to achieve a mAP value of 64.5 on the COCO test-dev [6] dataset.

Xue et. al [7] in their paper focus on litchi branch detection for robotic harvesting using semantic segmentation. It enhances DeepLabV3+ with an Xception-65 backbone, plus transfer learning and data augmentation to improve convergence and robustness. With an encoder-decoder design and ASPP and multi-rate atrous convolutions for small targets, they are able to achieve a mIoU of 0.765, outperforming the baseline DeepLabV3+ [8](0.621) and competing models like PSPNet-101 [9] and ICNet [10].

Menon et. al. in their work [11] use DeepLabV3+ to target semi-dark urban scenes by addressing receptive-field bias, limited representational power in residual encoders, and loss of spatial relationships from per-pixel labeling. The approach (“Unified DeepLabV3+”) combines custom atrous/dilated rates, non-linear shortcuts with group normalization, and superpixel-based (bunched) pixel cues to better preserve local-global context. On the CamVid [12] video dataset, it reports about a 3% gain in class-wise pixel accuracy over standard DeepLabV3+, and introduces an “ S^3 core” metric that fuses pixel accuracy, IoU, and boundary F-score for more robust evaluation.

In [13], work on UAV remote-sensing detection proposes Drone-DETR, an enhanced RT-DETR variant tailored for dense, small-object scenes in aerial imagery. It introduces a lightweight backbone (ESDNet) to retain shallow small-object cues and reduce redundant computation, and an attention-based module (EDF-FAM) with dual-path feature fusion for stronger multi-scale aggregation. By also using an extra P2 detection layer in the backbone, it significantly boosts small-object performance, reporting a mAP value of 53.9% on VisDrone2019 with 28.7M params—about an improvement of 8.1% over RT-DETR-R18.

III. DATA DESCRIPTION

The SPARK-2024 Spacecraft Dataset is meant to serve as a benchmark for spacecraft perception tasks. These include object detection and semantic segmentation, all based on satellite images taken in different visible conditions, such as variations in illumination or scale, viewpoint, and even how complex the background is. These features of this dataset make it very realistic and difficult to work with because it simulates the conditions of space observations as one would experience when using orbiters.

A standard supervised machine learning framework has been employed to divide the dataset into a training set, a validation set and a test set. The test set contains a collection

of images that have not been labelled and will only serve as the basis for evaluation and submission to the competition benchmark.

The dataset consists of ten (10) spacecraft categories: Cheops, LisaPathfinder, ObservationSat1, Proba2, Proba3, Proba3ocs, Smart1, Soho, VenusExpress and XMM Newton. There are large variations among the sizes and positions of the objects in this database (where images may have one or multiple contents, or no visible spacecraft present in the image).

The data is structured to enable reproducibility and portability by using the same directory structure within the data store for each image. As such, all images are stored using relative data paths to the root of the project. Preprocessing steps on the database include image annotation conversion, image normalization, image size adjustment, and data splitting. These steps will enable the combination of Detection and Segmentation Pipelines, as well as support reliable workflows between Training, Validation, and Inference.

Task	Training images	Validation images	Test images
Detection	60000	20000	20000
Segmentation	60000	20000	4000

TABLE I
DATASET SPLIT FOR BOTH TASKS

IV. METHODOLOGY

This project aims to develop a unified deep learning pipeline for spacecraft object detection and semantic segmentation using satellite imagery. The methodology is structured to ensure accuracy, scalability, and reproducibility throughout all stages of training, inference, and evaluation.

A. Data Preparation

1) *Annotation Conversion*: Annotation data from the dataset is transformed into COCO format to be used in the object detection task. This allows us to compute detection accuracy (Average Precision) across multiple Intersection over Union (IoU) thresholds (as defined in COCO evaluation). The COCO annotation format typically contains:

- Images: metadata for every image (width, height, file-name, etc.)
- Annotations: contains the bounding box (minX, minY, width, height)
- Categories: list of class definitions (id, name)

2) *Segmentation Mask Encoding*: The SPARK dataset has RGB images as the basis for segmentation ground-truth masks. For supervised training and quantitative evaluation purposes, the ground-truth mask data must be converted to an integer-valued label map format with the predefined class encoding below:

- Background: 0
- Body: 1
- Panel: 2

This same label scheme is used throughout the training models, generation of inference masks and for running evaluation scripts.

B. Spacecraft Detection

The objective of the Spacecraft Detection is to localize spacecraft instances and classify those spacecraft within satellite imagery into predefined categories. The Spacecraft Detection represents an implementation of a single stage object detector that directly predicts bounding boxes and class probabilities from input images.

1) *Detection Model and architecture:* To detect spacecraft, RT-DETRv2 is used, which is a transformer based object detection method that integrates convolutional feature extraction and global self-attention. For the process of detection, given an input image, it outputs a fixed number of object query detections for every input image and generates a class probability and bounding box for each of these detections.

$$\hat{y}_i = \left\{ \hat{c}_i, \hat{b}_i \right\}, \hat{b}_i = (c_x, b_y, w, h)$$

Where $\hat{c}_i, \hat{b}_i$ denote the initial box values, c_x, c_y the normalized box center and w, h its width and height.

This model was developed to maximize detection performance while providing good inference (or run-time) accuracy.

RT-DETRv2 has 3 primary components:

- **Backbone Network:** A multi-scale convolutional feature extractor that detects spacecrafts of diverse sizes and shapes.
- **Transformer-Based Decoder:** A query-based decoder that provides a pre-determined set of detections, each with class labels and localized central bounding boxes.

The overall design of RT-DETRv2 provides an optimal compromise between detection accuracy and computing efficiency that makes it suitable for the detection of large numbers of spacecrafts using high-performance computing platforms.

2) *Training setup:* The setup for training the detection model is based on configurations that are specified in YAML files so that they can be easily repeated for experimental traceability purposes. This approach facilitates re-training in the same controlled environment.

The RT-DETRv2 detector was optimized using an AdamW optimizer; this optimizer utilizes the ability of an adaptive learning rate to adjust automatically as the network converges, while the decoupled weight decay feature of AdamW helps with stabilizing the convergence rate of the network. Separate learning rates were used for both the backbone and detection head due to their differing optimization behaviors. The backbone was initialized from pre-trained weights, hence lower learning rates during training, while the detection head used higher learning rates during training so it could more rapidly adjust to the SPARK-2024 dataset.

In all training runs, a fixed number of training epochs is performed (40 epochs), with the model that achieves the greatest validation performance being selected for use in further inference and evaluation tasks. For this reason, the

Checkpoint file containing the highest Average Precision (AP) is used anytime the model is evaluated; this guarantees that any results obtained are always based on the model that performed best in validation.

The RT-DETRv2 framework's default preprocessing and augmentation strategies were followed during training. Each experiment used identical transformations (i.e., a set of uniform pre-processing operations including geometric transforms) that were defined in each configuration file.

Wandb has been used to monitor all training runs in real-time. This includes tracking the optimization behavior and validation performance as well as maintaining an auditable record of the training dynamics via visualizations.

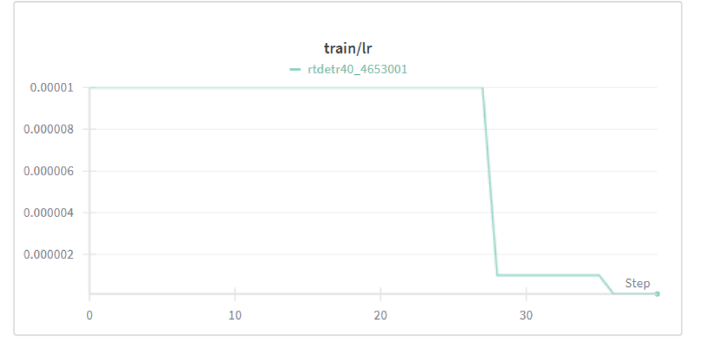


Fig. 1. Changes in learning rate over epochs



Fig. 2. Changes in loss over epochs

3) *Detection Evaluation:* **Intersection over Union (IoU)** was used to measure how well a predicted bounding box B_p corresponds to a true ground-truth bounding box B_g , we use an Intersection over Union (IoU) score.

$$IoU(B_p, B_g) = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$$

If the predictive class of the bounding box is the same as that of the corresponding ground truth bounding box and has a high enough IoU score, we classify this detection as correct.

Mean Average Precision (COCO) is determined by taking the average of all precision values across different IoU thresholds plotted on the precision-recall curve to give us a better sense of how well our model matches the accuracy

expected from the absolute points of bounding boxes in general. The average precision is given by:

$$AP = \frac{TruePositives}{TruePositives + FalsePositives}$$

We then compute the mAP for the AP values obtained at various thresholds for IoU.

$$mAP = \frac{1}{|T|} \sum_{t \in T} AP@t, \quad T = \{0.50, 0.55, \dots, 0.95\}$$

where T total number of threshold values taken and AP@t is the AP value at threshold value t.

Since AP considers both localization errors (i.e., bounding boxes not accurately placed) and misclassifications, it is the most common benchmark used to rate object detection performance.

Average Recall (AR) is based on the highest potential recall possible given an image with a defined object detection budget per image.

$$AR@k = \frac{1}{|T|} \sum_{t \in T} Recall@t$$

and Recall is given by:

$$R = \frac{TruePositives}{TruePositives + FalseNegatives}$$

C. Spacecraft segmentation

The objective of the spacecraft segmentation challenge is to produce pixel-level classifications for each input image to identify elements such as background or spacecraft components (spacecraft body and spacecraft solar panels). In the segmentation challenge for SPARK 2024, the task can be described as providing multiple pixel level segmentations for each input image (i.e., pixel level classification) for multi-class semantic segmentation.

1) *Segmentation Model*: DeepLabv3 is a convolutional neural network-based semantic segmentation model that generates dense pixel-level predictions. It uses atrous (dilated) convolution to provide multi-scale contextual information without losing spatial resolution, making it highly suitable for fine-grained satellite image segmentation.

The segmentation problem has been framed as a multi-class semantic segmentation task involving a background, spacecraft body, and solar panels. DeepLabv3 will take an input image and generate a class probability distribution for every pixel, creating the final segmentation mask by:

$$\hat{S}(x_p) = \arg_c \max P(c|x_p)$$

where x_p denotes a pixel input and $P(c|x_p)$ represents the predicted probability of class c.

By incorporating an Atrous Spatial Pyramid Pooling (ASPP) module with the backbone network of the convolutional net, DeepLabv3 can collect context from different locations (with different sizes of receptive fields) and does so in a very

efficient way that can handle the enormous differences in scale found in space. Additionally, it helps provide precise boundaries between the spacecraft body and its solar panels.

2) *Training setup*: The method for training the segmentation model is based on a patch-based approach; meaning that patches are randomly extracted from the original images at a resolution of 512x512 pixels per crop. Thus, this approach decreases the amount of memory used by the GPU and presents all patches with multiple spatial variations of each image type in a uniform resolution to all users.

Optimization of the model was performed using the AdamW optimization algorithm. Objective of which is based on calculating multi-class pixel-level Cross-Entropy loss, that considers the background, spacecraft body and solar panels, across a training sample by evaluating each pixel of the training image against the semantic class and then computing an overall mean loss for all pixels in the patch.

The model is trained until a predetermined number of epochs have been reached. There are model checkpoints saved and also a model checkpoint will be selected at the end of validation with the best overall validation performance, according to the criteria defined in the training script, to be used for inference/evaluation. All reporting will be with reference to the model checkpoint that had been available to the user prior to the training process beginning.

3) *Segmentation Evaluation*: Confusion matrices are used to evaluate segmentation performance at the pixel level; Intersection over Union (IoU) is calculated for each of the classes. The mean Intersection over Union (mIoU) value across all of the classes is the main quantitative measure of segmentation quality.

V. EXPERIMENTAL SETUP

All tests were performed on a GPU based high performance computing cluster built on Nvidia V100 GPUs, that used SLURM to manage resources. A Python based pipeline that used PyTorch was created with separate environments for detection and segmentation.

Both RT-DETRv2 and DeepLabv3 were trained through supervised learning using COCO formatted annotations and pixel-wise masks, respectively. The inference process used GPU acceleration to generate structured detections and pixel based segmentations. All experiments were tracked and made reproducible by using WandB.

VI. RESULTS AND ANALYSIS

In this section, we will describe the quantitative and qualitative results achieved for detecting spacecraft and performing semantic segmentation using the SPARK-2024 dataset. The performance of each of the four methods evaluated in the introduction was assessed by applying them to the test split and submitting the results for benchmarking purposes.

A. Spacecraft Detection results

1) *Quantitative Detection Performance* : COCO-style detection metrics are utilized to evaluate the RT-DETRv2 detector performance based on various intersection over union

(IoU) thresholds. The evaluation uses precision and recall over different detection constraints.

The performance of the proposed detection model according to standard COCO metrics is consistent and very good, as shown in Table II. The average precision through all IoU thresholds (ranging between 0.50 - 0.95) is calculated to be 0.969, it demonstrates excellent localisation. The average precision of 0.997 at IoU = 0.50 demonstrates an almost perfect localisation accuracy.

Metric	Value
AP@0.50	0.997
AP@0.75	0.994
mAP (0.50 to 0.95)	0.969
AR@100	0.978

TABLE II
DETECTION PERFORMANCE METRICS

At the stricter localisation threshold of IoU = 0.75, the performance remains very good with an average precision of 0.994, indicating a high accuracy in bounding box regression. The performance on larger scale objects is excellent with an average precision of 0.970, but the performance on medium scale objects was lower with an average precision of 0.736. The lower performance on medium scale objects highlights the challenges of scale variability and the difficulty it presents in obtaining accurate results when analysing spacecraft imagery. Using standard COCO model (AR@100), the detector has a high average recall of 0.978, which shows that it consistently detects objects throughout the images. Metrics for small objects were not available due to a lack of small instances in the dataset.

2) *Detection Stability and Confidence Threshold Analysis:* Using a threshold sweep performed upon the validation set, the resilience of the detector to the confidence threshold was evaluated. The model produces approximately 5002 detections over 5000 images at operating point ($\tau = 0.30$) which correspond to the nature of the dataset where there is an expected average of one detection per image.

The count of detections remains stable at threshold levels of 0.05 through 0.40, and slightly decreases as threshold levels increase above 0.40, which indicates that the score for confidence is well-calibrated. The retained detections from the validation set possess relatively high scores for confidence ($mean \approx 0.97$), which indicates that there is a clear distinction between the foreground objects and the background objects.

Together, it can be noted that the detector will maintain stable inference performance and was therefore demonstrated to be appropriate at the operating point of $\tau = 0.30$ for evaluation and benchmark submissions.

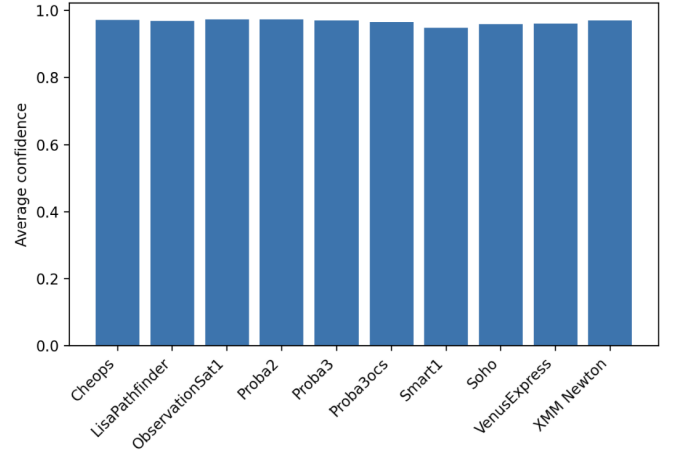


Fig. 3. Confidence per class (thr = 0.30)

We also assessed the performance of each spacecraft class by calculating average "COCO" styled precision across various intersections 0.5–0.95; these are shown in Table III; upon evaluation it was determined that, on average, every type of spacecraft achieved an AP score greater than 95%, therefore indicating consistency in performance.

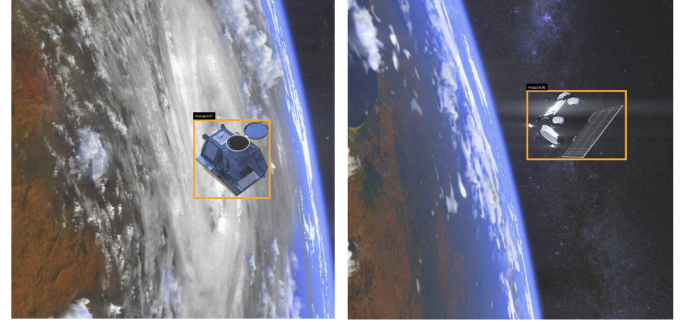


Fig. 4. Bounding boxes for Object detection task

The type of spacecraft with the highest average precision (Proba2) achieved an AP score of 0.986, whereas Smart1 obtained an AP of 0.936 due to intra-class variance and visual similarity between some spacecraft. Importantly, the overall performance for no single category showed substantial degradation, demonstrating balanced generalisation and no class-specific overfitting.

Class	mAP
Proba2	0.9861
Cheops	0.9838
ObservationSat1	0.9813
LisaPathfinder	0.9810
Proba3	0.9801
XMM Newton	0.9743
Proba3docs	0.9619
VenusExpress	0.9559
Soho	0.9529
Smart1	0.9360

TABLE III
PER CLASS DETECTION PERFORMANCE

B. Spacecraft Segmentation Results

1) *Dataset Characteristics and Pixel Imbalance*: The spacecraft segmentation task exhibits extreme pixel-level class imbalance. The validation set is dominated by pixels representing the background, with very few that represent the body of the spacecraft and the solar panels. Specifically, the solar panel component makes up about 1.2% of the total number of pixels in the image, rendering them very vulnerable to boundary errors and several forms of misclassification due to their low pixel count.

Class	Pixels ($\times 10^6$)	Percentage (%)
Background	1996.24	95.21
Body	75.90	3.62
Panels	25.01	1.19
Total	2097.15	100.00

TABLE IV
PER CLASS DETECTION PERFORMANCE

Thus, the pixel-count imbalance leads to significant increases in difficulty with the semantic segmentation of thin structures such as solar panels, where small absolute pixel misclassification errors can generate substantial changes in Intersection-over-Union (IoU) values.

2) *Quantitative Segmentation Performance*: Table V summarizes the evaluation for segmentation performance using pixel-by-pixel IoU for each class. Since the dataset had a high percentage of background samples compared to foregrounds, the IoU values for the background class are much higher than the other classes. In addition, the presence of fine-grained structures made it difficult to achieve high foreground IoUs, resulting in very low IoU values for all classes except for the background.

Class	IoU
Background	0.967
Body	0.003
Panels	0.009
mIoU	0.326

TABLE V
SEGMENTATION IOU OVER CLASS

The low values for IoU in the foreground confirm what is known about the type of data contained in the specific datasets; therefore, this is consistent with the information available on pixel-perfect segmentation of very small and/or slender shapes.

Thus, metrics that use quantifiable statistical numbers alone will not show how well the actual segmentation system works in practice.

To obtain a more accurate picture of the actual segmentation system performance, it is important to also evaluate only the confusion in the foreground.

3) *Qualitative Segmentation Analysis*: In the case of extreme class imbalance, it proves to be inefficient to perform a pixel-level IoU evaluation. Therefore, to assess the segmentation quality, particular emphasis is placed on qualitative analysis through visual checks of the overlay of the

predictions. These checks indicate that the network does an excellent job in identifying the spacecraft outline, though the solar panels are still the hardest part to detect.

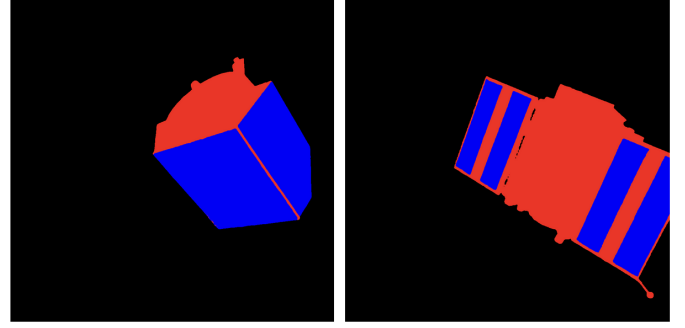


Fig. 5. Semantic segmentation masks for the spacecraft

The errors tend to be concentrated in regions of thin boundaries, low-contrast areas, and small-scale panel representation. Notwithstanding the difficulties, the prediction is spatially consistent.

4) *Summary of Segmentation Findings*: In short, segmentation is far more difficult compared to detection because of extreme pixel imbalance and fine structural details. Even though the IoU metrics quantitatively for the foreground classes are low, the qualitative results verify that the meaningful spacecraft regions have been well captured by the model. These results point towards improvements in future work: forensic loss functions that are aware of the foreground, refinement of boundaries, and panel-specific modeling.

VII. CONCLUSION

The current research focused on the task of spacecraft detection, as well as semantic segmentation, using the SPARK-2024 dataset, with the help of deep learning models. For spacecraft detection, the architecture of RT-DETRv2 demonstrated very good and stable results, with high COCO validation metrics, independent of the threshold confidence level. The experimental results are very close to the competitive performance achieved on the Codabench benchmark.

The results for semantic segmentation indicate that the process became much more complicated owing to the large class imbalance on a pixel level and the existence of thin structures like solar panels on the spacecraft. Despite a low IoU score concerning the foreground, it can be noted that the model accurately picked up the spacecraft silhouette and main body parts.

REFERENCES

- [1] Y.-H. Tseng and S.-S. Jan, "Combination of computer vision detection and segmentation for autonomous driving," in *2018 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2018, pp. 1047–1052. DOI: 10.1109/PLANS.2018.8373485.

- [2] W. Lu, J. Chen, and F. Xue, "Using computer vision to recognize composition of construction waste mixtures: A semantic segmentation approach," *Resources, Conservation and Recycling*, vol. 178, p. 106022, 2022, ISSN: 0921-3449. DOI: <https://doi.org/10.1016/j.resconrec.2021.106022>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921344921006303>.
- [3] Q. Chen et al., *Group detr v2: Strong object detector with encoder-decoder pretraining*, 2022. arXiv: 2211.03594 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2211.03594>.
- [4] Q. Chen et al., *Group detr: Fast detr training with group-wise one-to-many assignment*, 2023. arXiv: 2207.13085 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2207.13085>.
- [5] H. Zhang et al., *Dino: Detr with improved denoising anchor boxes for end-to-end object detection*, 2022. arXiv: 2203.03605 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2203.03605>.
- [6] T.-Y. Lin et al., *Microsoft coco: Common objects in context*, 2015. arXiv: 1405.0312 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1405.0312>.
- [7] H. Peng et al., "Semantic segmentation of litchi branches using deeplabv3+ model," *IEEE Access*, vol. 8, pp. 164 546–164 555, 2020. DOI: 10.1109/ACCESS.2020.3021739.
- [8] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, *Encoder-decoder with atrous separable convolution for semantic image segmentation*, 2018. arXiv: 1802.02611 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1802.02611>.
- [9] Z. Zhao, Y. Mao, Y. Ding, P. Ren, and N. Zheng, "Visual-based semantic slam with landmarks for large-scale outdoor environment," in *2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI)*, 2019, pp. 149–154. DOI: 10.1109/CCHI.2019.8901910.
- [10] H. Zhao, X. Qi, X. Shen, J. Shi, and J. Jia, *lcnets for real-time semantic segmentation on high-resolution images*, 2018. arXiv: 1704.08545 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1704.08545>.
- [11] M. M. Memon, M. A. Hashmani, A. Z. Junejo, S. S. Rizvi, and K. Raza, "Unified deeplabv3+ for semi-dark image semantic segmentation," *Sensors*, vol. 22, no. 14, p. 5312, 2022.
- [12] G. J. Brostow, J. Shotton, J. Fauqueur, and R. Cipolla, "Segmentation and recognition using structure from motion point clouds," in *ECCV (1)*, 2008, pp. 44–57.
- [13] Y. Kong, X. Shang, and S. Jia, "Drone-detr: Efficient small object detection for remote sensing image using enhanced rt-detr model," *Sensors*, vol. 24, no. 17, p. 5496, 2024.